

Chemical Equilibrium

Module 6 · 2105603 Advanced Chemical Engineering Thermodynamics

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After-midterm classroom sequence

This module remains the extended reference deck. The timed teaching sequence uses two sessions with guided labs:

- Session 5 slides · Lab tasks and independent practice
- Session 6 slides · Lab tasks and independent practice

Each class is 180 minutes including a break, demonstration and guided exploration. Additional theory in this reference deck is optional reading.

Nothing new is introduced in this module.

The same criterion, the same fugacity coefficient, the same activity coefficient — under a different constraint.

Where this module ends

Everything from Module 1 onward feeds one constrained optimisation, and that optimisation is what an equilibrium reactor block in a process simulator solves.

The arc

Module 1 gave P - V - T . Module 2 turned it into fugacity. Module 3 gave the liquid an excess Gibbs energy. Module 4 fitted it to data. Module 5 asked whether the answer was stable.

Module 6 minimises the total Gibbs energy of everything at once.

Four questions

- What is the equilibrium criterion for a reaction, and why is it the same one?
- Where do non-ideality corrections enter, and how much are they worth?
- What do you do when nobody knows the reactions?
- What happens when phases and reactions must be solved together?

The criterion

Section 6.1 Extent of reaction, $\sum_i \nu_i \mu_i = 0$, and the sharp distinction between K and composition.

Extent of reaction

Stoichiometry removes almost all of the freedom. One number describes the whole composition of a single reaction.

$$n_i = n_{i0} + \nu_i \xi \quad \Longrightarrow \quad \frac{dn_i}{\nu_i} = d\xi \text{ for every } i$$

Why it is one number

Atoms are conserved, so the composition cannot move in an arbitrary direction. For a single reaction it can move along exactly one line in composition space, and ξ is the distance along it.

The bounds matter

ξ is limited at one end by the limiting reactant and at the other by the products. Outside that window a mole number is negative and the objective evaluates $\ln$ of a negative number. A solver started outside it fails in a way that looks like a physics problem and is not.

The equilibrium condition

At constant T and P , $dG = 0$. Substituting the extent gives the whole of reaction equilibrium in one line.

$$dG = \sum_i \mu_i dn_i = \left(\sum_i \nu_i \mu_i \right) d\xi = 0 \quad \Rightarrow \quad \boxed{\sum_i \nu_i \mu_i = 0}$$

The same statement

Phase equilibrium says $\mu_i^\alpha = \mu_i^\beta$ — a *transfer* of species i between phases leaves G unchanged. Reaction equilibrium says a *conversion* along the stoichiometry leaves G unchanged. Both are $dG = 0$; only the allowed variation differs.

What follows

Substituting $\mu_i = G_i^\circ + RT \ln \hat{a}_i$ and rearranging gives

$$\prod_i \hat{a}_i^{\nu_i} = \exp\left(-\frac{\Delta G^\circ}{RT}\right) = K$$

with $\Delta G^\circ = \sum_i \nu_i G_i^\circ$.

K depends only on temperature

The single most-confused point in the subject. The composition depends on pressure and on non-ideality; K does not.

Why

$K = \exp(-\Delta G^\circ / RT)$ and ΔG° is built from *standard-state* properties. The standard state fixes the pressure — 1 bar for a gas — so nothing on the right-hand side knows the system pressure.

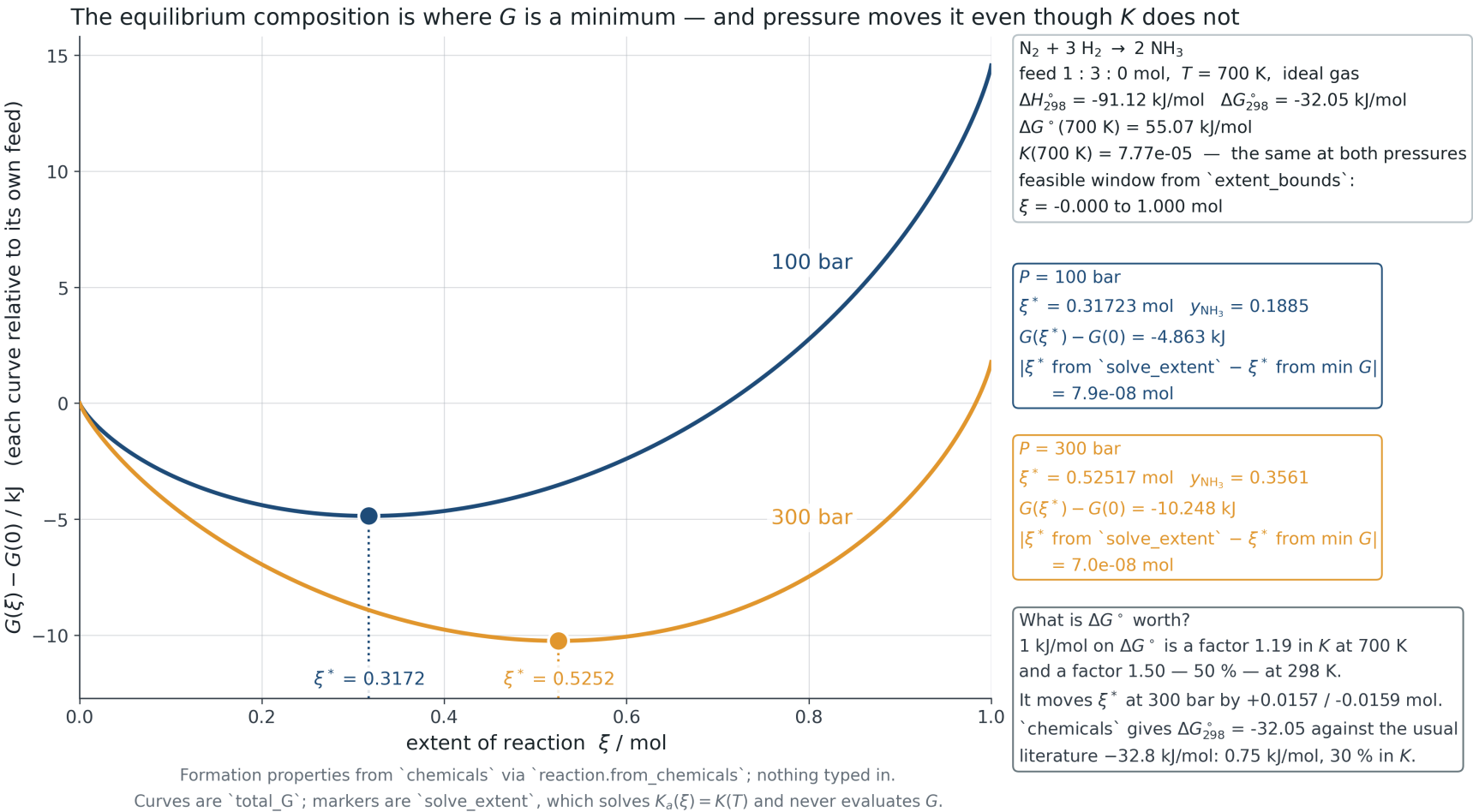
What does move

The activity quotient. For a gas $\prod \hat{a}_i^{\nu_i} = K_\phi K_y (P/P^\circ)^{\Delta\nu}$, so raising P with $\Delta\nu < 0$ forces K_y down — more product — at constant K .

Verified, not asserted

For ammonia synthesis at 700 K, raising the pressure from 1 to 300 bar changes K by exactly nothing and multiplies the ammonia mole fraction by more than a hundred.

Gibbs energy against extent



F6.1 • The minimum moves with pressure. K does not.

Temperature dependence

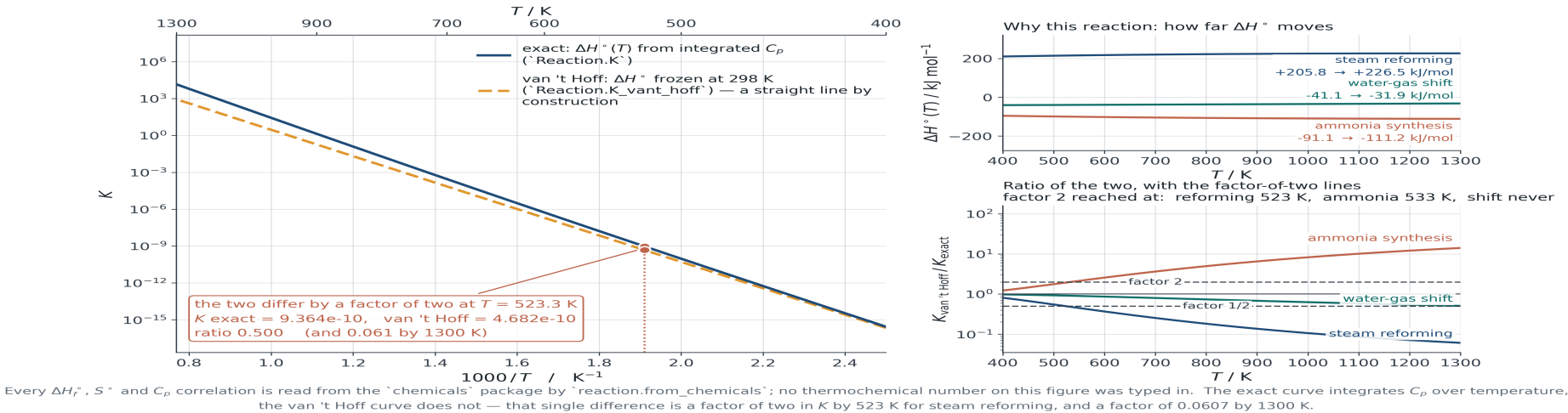
Van 't Hoff, and where the constant-enthalpy shortcut stops being a shortcut.

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

$\Delta H^\circ \text{ constant}$ $\longrightarrow$

$$\ln \frac{K(T)}{K(T_0)} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right)$$

The straight line of the van 't Hoff plot is an assumption about ΔH° , and it is worth a factor of two in K well inside the range this course uses
Steam methane reforming: $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3 \text{H}_2$



F6.2 • Exact against constant-enthalpy, with the temperature at which they differ by a factor of two.

Standard states, and the traps in each

Gas

Pure ideal gas at 1 bar. $\hat{a}_i = \hat{\varphi}_i y_i P / P^\circ$. The trap: forgetting P° , which makes K dimensional and wrong by $(P/P^\circ)^{\Delta\nu}$

Liquid

Pure liquid at T and P . $\hat{a}_i = \gamma_i x_i$. The trap: the Lewis-Randall reference is the *pure* liquid, which may not exist at those conditions

Solid

Pure solid. $\hat{a}_i = 1$, so it drops out of K entirely — but not out of the question of whether the solid is present at all

Solute

Hypothetical ideal 1 molal solution. $\hat{a}_i = \gamma_i m_i / m^\circ$. The trap: molarity, molality and mole fraction give three different numerical K values for the same physics

Non-ideality

Section 6.2 Where $\hat{\varphi}_i$ from Module 2 and γ_i from Module 3 re-enter.
Nothing new is introduced. Existing tools meet a new constraint.

The decomposition

Split the activity quotient into a composition part and a non-ideality part. Each factor is something you already know how to compute.

$$K = \underbrace{\prod_i \hat{\varphi}_i^{\nu_i}}_{K_\varphi} \underbrace{\prod_i y_i^{\nu_i}}_{K_y} \left(\frac{P}{P^\circ} \right)^{\Delta\nu} \quad (\text{gas})$$

$$K = \underbrace{\prod_i \gamma_i^{\nu_i}}_{K_\gamma} \underbrace{\prod_i x_i^{\nu_i}}_{K_x} \quad (\text{liquid})$$

K_φ is a ratio

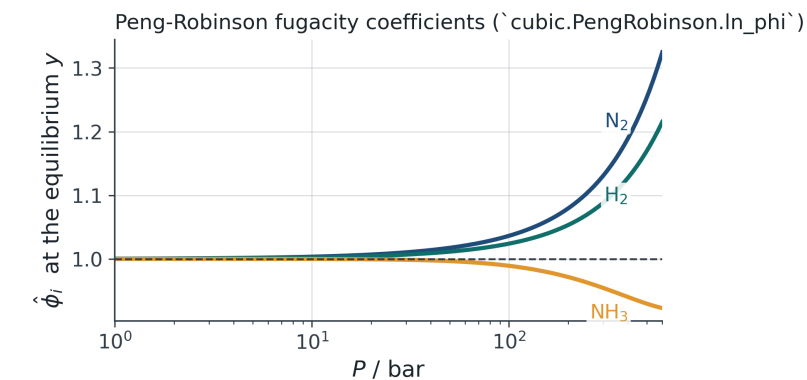
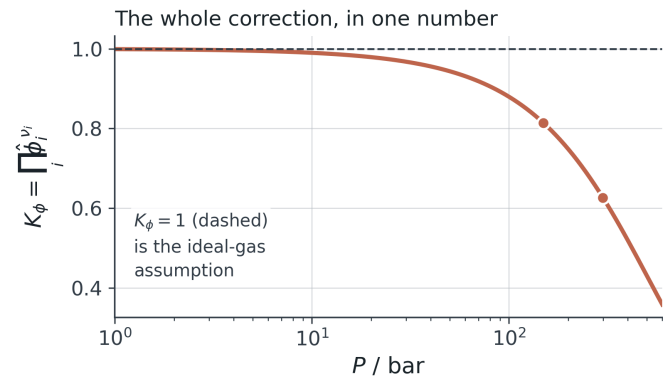
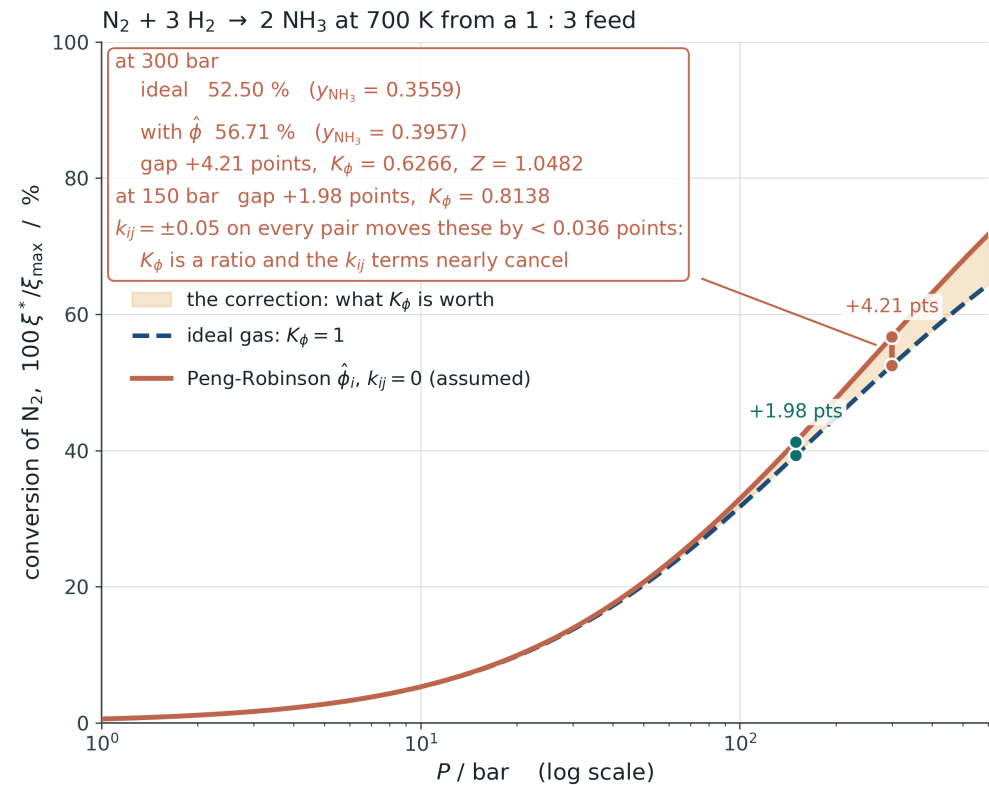
Which has a useful consequence: much of the model error cancels. Changing every binary interaction parameter by ± 0.05 moves the ammonia conversion by less than 0.04 percentage points at any pressure marked on the next slide. The fugacity coefficients themselves are worth 4.2 points at 300 bar.

Where the effort belongs

At 300 bar the corrections rank: $\hat{\varphi}$ is worth 4.2 points of conversion, an uncertainty of 1 kJ/mol in ΔG° is worth 1.6 points, and k_{ij} is worth less than 0.04 — forty times smaller than the thermochemical uncertainty. Improve the data before the mixing rule.

Ammonia synthesis, done properly

At 300 bar the fugacity coefficient is not a refinement: it is +4.2 percentage points of conversion



ASSUMED, NOT FITTED: $k_{ij} = 0$ for N_2/H_2 , N_2/NH_3 and H_2/NH_3 . No binary VLE data were regressed here, and none was found in a citable form. Recomputing with $k_{ij} = \pm 0.05$ on every pair changes nothing visible. T_c , P_c , ω from ``chemicals``. Hydrogen sits at $T_r = 21$ here, far outside the range the Peng-Robinson $m(\omega)$ polynomial was fitted in — a known weakness, stated rather than corrected. For scale: ± 1 kJ/mol on ΔG° moves the 300 bar answer to 58.3 / 55.1 % against 56.7 %.

F6.3 • Conversion against pressure with and without K_ϕ , from Peng-Robinson. $k_{ij} = 0$ assumed and labelled.

Reactions in solution

Three concentration scales, three numerical values of K , one physical answer.

Mole fraction

Lewis-Randall reference: $\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$. Natural for a solvent and for a mixture of comparable components.

Molality

Henry reference: $\gamma_i \rightarrow 1$ as $m_i \rightarrow 0$. Natural for a solute that never approaches purity, and the scale on which electrolyte data are tabulated.

Molarity

Convenient in the laboratory and awkward in thermodynamics, because it depends on the solution density and therefore on temperature and on composition.

Converting between scales changes K and ΔG° together. A reported K without its concentration scale is not a number, and neither is a ΔG° without its standard state.

When nobody knows the reactions

Section 6.3 Multiple extents, and then the formulation that needs no reaction set at all.

Several reactions at once

The extent formulation generalises directly — as long as the reaction set is known and independent.

$$n_i = n_{i0} + \sum_j \nu_{ij} \xi_j$$

$$\sum_i \nu_{ij} \mu_i = 0 \quad \text{for each } j$$

How many are independent

$R = S - \text{rank}(\mathbf{A})$, with S species and $\mathbf{A}$ the element matrix. Gas-phase steam reforming: 5 species, 3 elements, rank 3, so **2** independent reactions — and which two you write down is a choice, not a fact. Admit solid carbon and it becomes 6 species and 3 reactions.

Why the choice does not matter

Any independent set spans the same reaction space and gives the same equilibrium. If two choices give different answers, one set was not independent — which is worth checking with a rank rather than by inspection.

Where the extent formulation fails

It needs a reaction set. Combustion, reforming and pyrolysis do not come with one.

The problem

A hydrocarbon flame contains hundreds of species. Nobody writes down the reaction set; there is no agreed set to write. Any list you choose is a modelling assumption you cannot defend, and omitting a species that matters is invisible in the result.

The alternative

Minimise the total Gibbs energy over all species subject to conservation of atoms. No stoichiometry is required, only an element list — and an element list is not a modelling choice.

$$\min_{\mathbf{n}} \sum_i n_i \mu_i \quad \text{s.t.} \quad \mathbf{A}\mathbf{n} = \mathbf{b}, \quad n_i \geq 0$$

Element potentials

The Lagrange multipliers of the element balances are not bookkeeping. They are the chemical potential of an atom.

$$\mathcal{L} = \sum_i n_i \mu_i - \sum_k \lambda_k \left(\sum_i a_{ki} n_i - b_k \right) \implies \boxed{\mu_i = \sum_k a_{ki} \lambda_k}$$

What it says

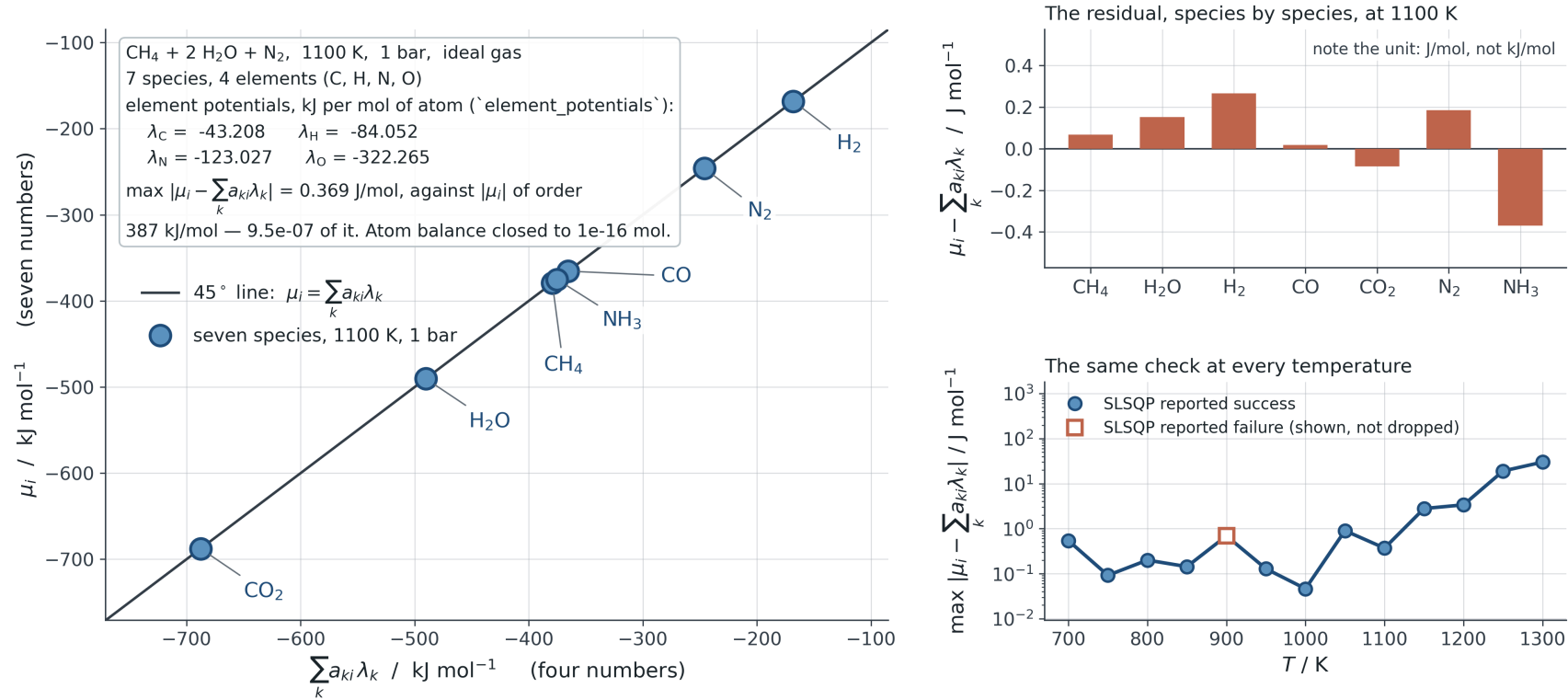
At equilibrium a species carries no information beyond the atoms it is made of. Its chemical potential is a weighted sum of element potentials, and the weights are its formula.

Why it is useful

It is the cheapest possible check on a converged solution: the residual of that relation costs nothing to compute and does not require re-solving anything. For the reforming case it comes out at 0.37 J/mol against chemical potentials of order 400 kJ/mol.

What the multipliers look like

Seven chemical potentials, four numbers: at equilibrium a species carries no information beyond the atoms it is made of



Formation properties from 'chemicals'. The minimisation is 'reaction.gibbs_minimise' (SLSQP on $\ln n$ subject to $A n = b$); the multipliers are the least-squares solution of $A^T \lambda = \mu$ and are never used to obtain the composition, so the residual above is an independent check. O₂ and NO were removed from the species list: their equilibrium mole fractions here are of order 10^{-25} and 10^{-8} , and the minimiser cannot resolve $\ln n$ for a species that contributes nothing to G .
Equilibrium composition at 1100 K, 1 bar: CH₄ 0.00046471, H₂O 0.13416, H₂ 0.53216, CO 0.13323, CO₂ 0.033135, N₂ 0.16681, NH₃ 4.8673e-05.

F6.4 • Every species on the 45° line. The residual, and the one temperature at which the optimiser failed, are both reported.

The numerical difficulty

The objective contains $\sum n_i \ln(n_i/n)$, which is ill-conditioned as any mole number approaches zero. Student code fails here, and so does ours.

What goes wrong

A trace species contributes almost nothing to G and has a large derivative. Finite-difference gradients lose it entirely. Below about $y = 10^{-10}$ the solution is not resolved — adding O_2 and NO to a reforming calculation drove the element-potential residual from 0.37 to 5×10^4 J/mol.

What helps

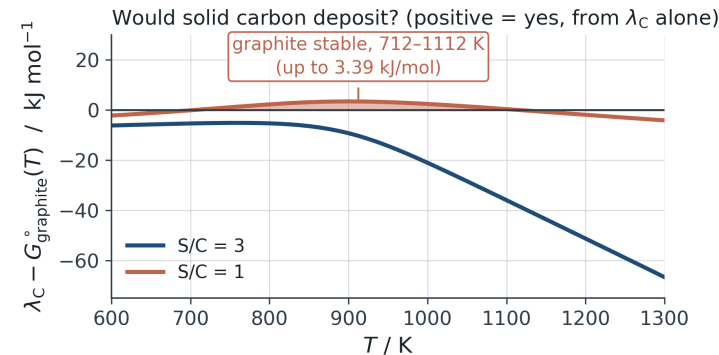
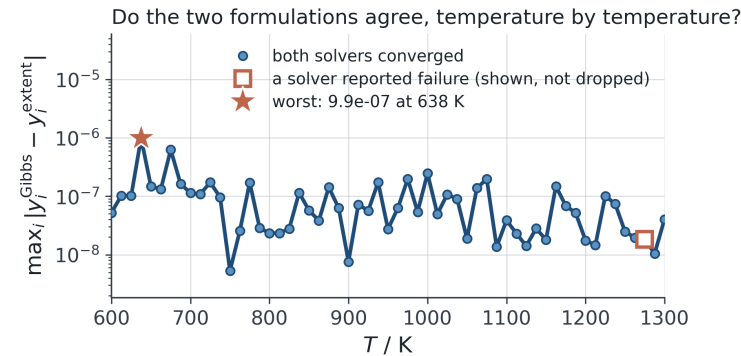
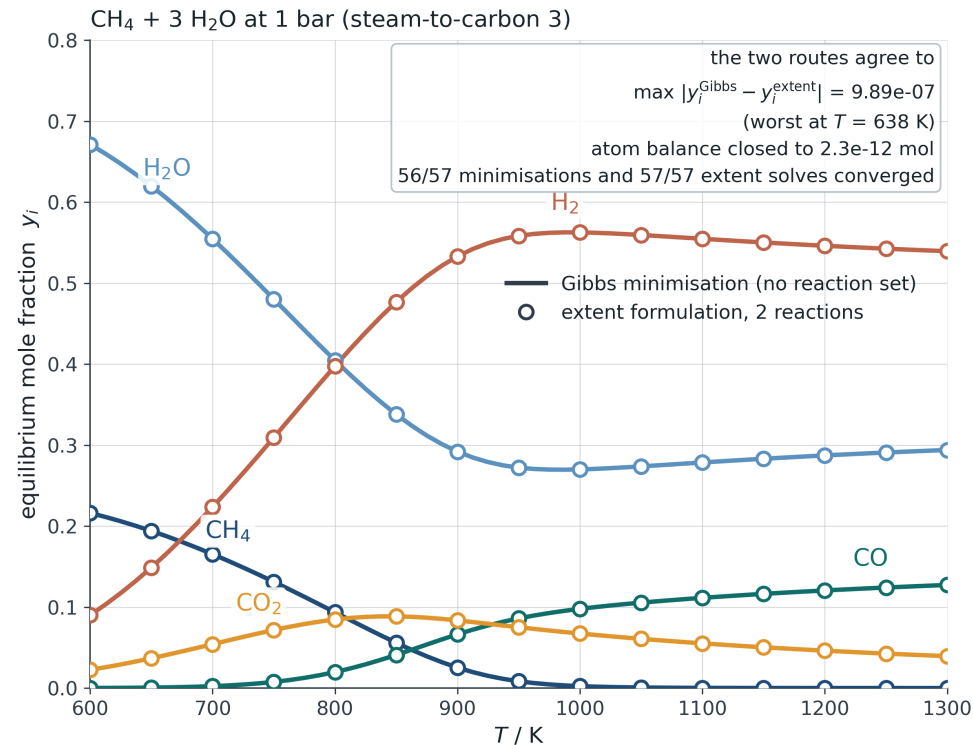
Optimise in $\ln n_i$, which enforces positivity exactly rather than by a constraint the solver can step across. Restart from several compositions. Drop species that cannot matter — and say that you dropped them.

What settles it

Cross-check against the extent formulation wherever a reaction set *is* known. For steam reforming over 600–1300 K the two agree to 10^{-6} in mole fraction. Build that check in before you trust an answer you cannot verify.

Product distribution from Gibbs minimisation

Two formulations of the same problem, five species and three elements, agreeing to 9.9×10^{-7} in mole fraction at every temperature



All formation properties from `chemicals`; graphite from `Hfs`/'S0s` for CAS 7782-42-5 with the JANAF solid C_p table. The check that graphite is right: $\text{C} + \text{CO}_2 \rightarrow 2 \text{CO}$ gives $\Delta H_{298}^\circ = 172.4$, $\Delta G_{298}^\circ = 120.0 \text{ kJ/mol}$ (literature 172.5, 120.0) and $\text{CH}_4 \rightarrow \text{C} + 2 \text{H}_2$ gives 74.5, 50.4 kJ/mol (literature 74.9, 50.8). The main panel is a single gas phase: at $\text{S/C} = 3$ graphite is never stable, so that is the true equilibrium; at $\text{S/C} = 1$ it is not, inside the shaded band.

Phases, reactions and charge

Section 6.4 Everything at once, and then the one place where a potential difference enters the chemical potential itself.

Reactive flash

Nothing changes. The species list gains a phase label, the element balance sums over both phases, and the same minimisation runs.

$$\min_{\{n_i^\alpha\}} \sum_{\alpha} \sum_i n_i^\alpha \mu_i^\alpha \quad \text{s.t.} \quad \sum_{\alpha} \mathbf{A} \mathbf{n}^\alpha = \mathbf{b}$$

Why this is the ending

Phase equilibrium falls out as the condition $\mu_i^\alpha = \mu_i^\beta$; reaction equilibrium falls out as $\sum \nu_i \mu_i = 0$. Neither was imposed. Both are consequences of minimising one function subject to atom conservation.

And the stability question returns

The minimisation is only over the phases you listed. Whether *another* phase should be present is the Module 5 question, and the answer is still the tangent plane criterion. A reactive flash that was never asked about a second liquid will not find one.

Reactive distillation

Why it works, not how it is designed.

The trick

Run the reaction inside the column. Removing a product by distillation as it forms pulls the reaction past its equilibrium conversion — the composition never sits at equilibrium, because the separation keeps moving it.

Equilibrium-limited esterifications are the standard case: methyl acetate production replaced a reactor and eight columns with one column.

Why it is hard

The reaction changes the phase behaviour and the phase behaviour changes the reaction. Reactive azeotropes exist that are not azeotropes of the non-reacting mixture, and the distillation boundaries of Module 5 move.

Nothing about the design is separable, which is why it is a specialist subject and why the thermodynamics has to be right first.

Electrolytes

Long-range Coulomb interactions make an electrolyte non-ideal at concentrations where a molecular solution would still be ideal.

$$\log_{10} \gamma_{\pm} = -A|z_+ z_-| \sqrt{I}$$

$$\log_{10} \gamma_{\pm} = \frac{-A|z_+ z_-| \sqrt{I}}{1 + Ba\sqrt{I}}$$

$$I = \frac{1}{2} \sum_i m_i z_i^2$$

Limiting law

Derived, with no fitted parameter, from the Poisson-Boltzmann equation and a point-charge ion. Exact as $I \rightarrow 0$ and 1 % wrong by $I = 0.005$ mol/kg.

Extended law

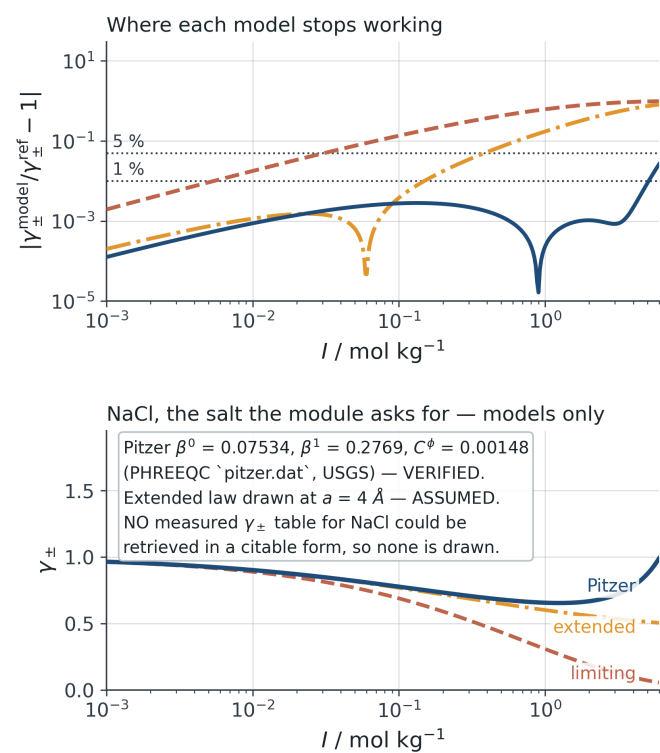
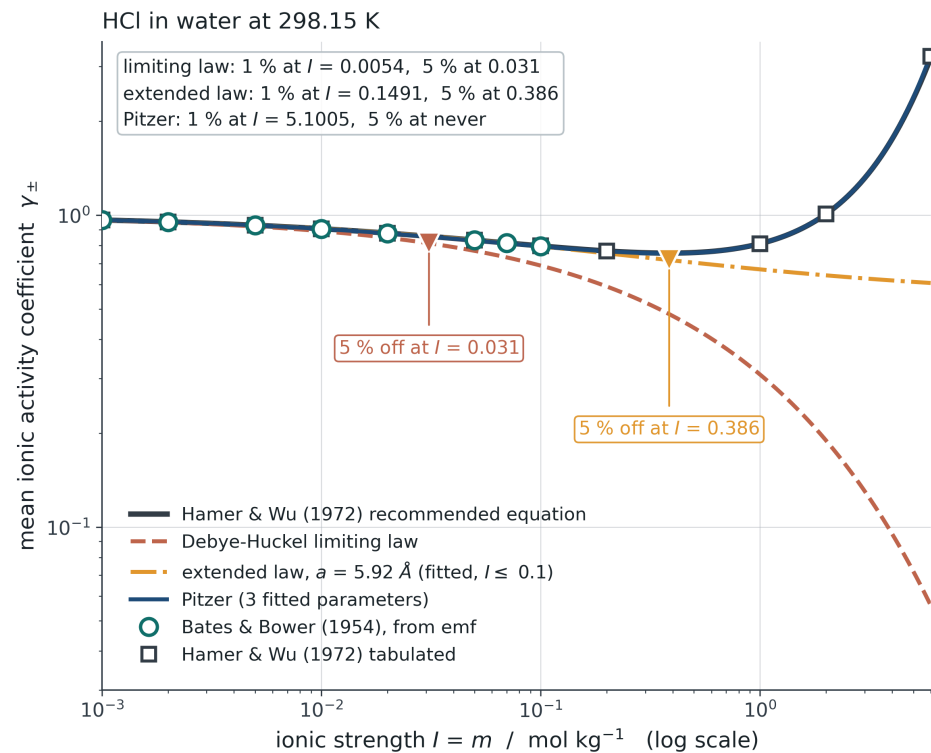
One fitted ion-size parameter. Buys a factor of thirty: 1 % wrong by $I = 0.15$ mol/kg.

Pitzer

Three fitted parameters and a virial-like expansion. 1 % to about $I = 5$ mol/kg — which is where industrial brines and battery electrolytes actually sit.

How far each one reaches

The limiting law is finished by $I = 0.03$ mol/kg and the extended law by 0.39; three fitted Pitzer parameters carry the same solution to 6 mol/kg



Reference $\gamma_{\pm}$: Hamer & Wu, J. Phys. Chem. Ref. Data 1, 1047 (1972), Table 4 (equation and tabulated values), cross-checked against Bates & Bower, J. Res. NBS 53, 283 (1954), Table 3 — the two agree to 0.0012 in $\gamma_{\pm}$.
Pitzer parameters from PHREEQC 'pitzer.dat' (USGS). Debye-Huckel $A = 0.5108$ from 'reaction.debye_huckel_A', against the 0.5108 Hamer & Wu used. ONE FLAG: the retrieved table value at $m = 6$ (3.23) sits +1.1 % from the retrieved equation (3.196); every other point agrees to 0.06 %, so that one figure is most likely a retrieval error and is shown rather than quietly corrected.

F6.6 • Three models against critically-evaluated data for HCl(aq) at 298.15 K. The electrolyte is HCl rather than NaCl because that is the data that could be retrieved and corroborated.

Electrochemical potential

The one place in this course where a potential difference enters the chemical potential itself.

$$\tilde{\mu}_i = \mu_i + z_i F \phi \qquad \sum_i \nu_i \tilde{\mu}_i = 0 \qquad \implies \qquad E = E^\circ - \frac{RT}{nF} \ln Q$$

Derived, not quoted

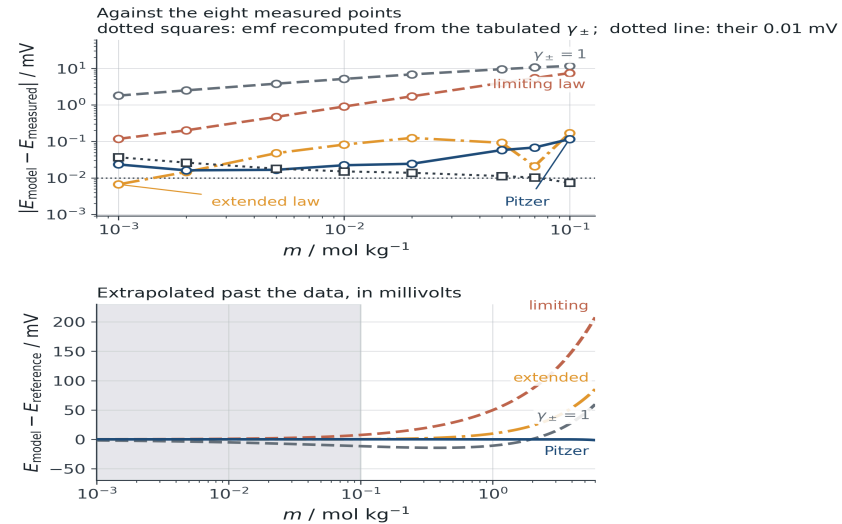
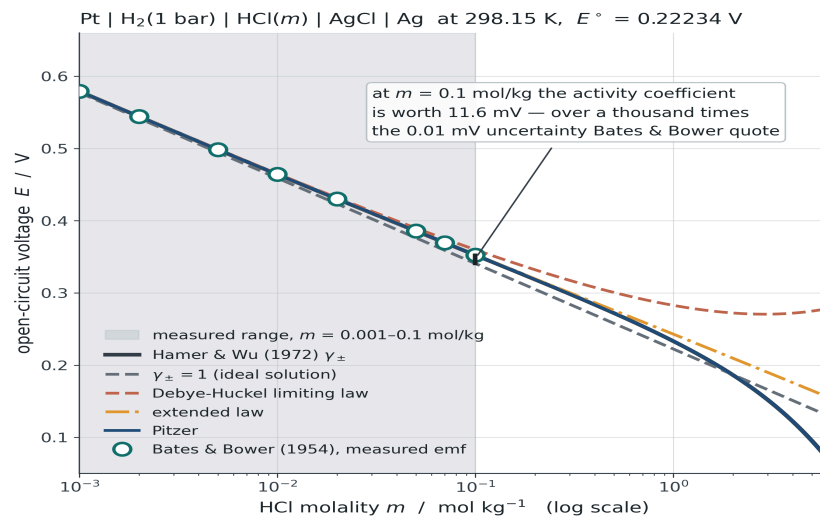
Nernst is not a new principle. It is $\sum_i \nu_i \tilde{\mu}_i = 0$ with the electrical term collected: the $z_i F \phi$ contributions across the two electrodes sum to $nF(\phi^{\text{R}} - \phi^{\text{L}}) = nFE$, and the rest is the activity quotient.

The slope is a check

$2.303 RT/F = 59.16$ mV per decade at 25 °C. If your derivation gives anything else, the electron count or a sign is wrong — and it is checkable before any data are involved.

What the activity coefficient is worth, in millivolts

The activity coefficient is not a correction to a calculation — it is 12 mV on a voltmeter at 0.1 mol/kg, and 60 mV at 6



MEASURED, NOT ILLUSTRATIVE. E° and the eight emf values are Bates & Bower, J. Res. NBS 53, 283 (1954), Tables 2 and 3; the reference $\gamma_{\pm}$ curve is Hamer & Wu, J. Phys. Chem. Ref. Data 1, 1047 (1972), Table 4. Pitzer parameters from PHREEQC `pitzer.dat` (USGS). The extended-law ion size $a = 5.92$ Å is fitted here to the measured $\gamma_{\pm}$ below 0.1 mol/kg. Recomputing the measured emf from the measured $\gamma_{\pm}$ through the Nernst equation closes to 0.04 mV, which is the internal consistency of the data set and the floor for every model comparison above. Beyond 0.1 mol/kg the curves are predictions with nothing to test them against, and the shaded band marks where the data end.

The number

Against the Bates and Bower cell data, the rms error is 7.4 mV if $\gamma_{\pm}$ is taken as one, 3.6 mV with the limiting law, 0.09 mV with the extended law, and 0.054 mV with Pitzer.

Why it matters

At 0.1 molal the activity coefficient is worth 11.6 mV, against a measurement uncertainty of 0.01 mV. A battery state-of-charge estimate built on ideal activities is wrong by a thousand times the resolution of its own sensor.

What you must be able to do

The module in six statements.

1 Derive the equilibrium condition.

From $dG = 0$ to $\sum \nu_i \mu_i = 0$ to K , and say why it is the same criterion as phase equilibrium.

2 Separate K from composition.

K is a function of temperature alone; everything else moves with pressure and non-ideality.

3 Put the corrections in the right order.

$\hat{\phi}$, then ΔG° uncertainty, then k_{ij} — and know the size of each.

4 Minimise Gibbs energy with an element balance.

Set up the constraints, interpret the multipliers, and know where the objective is ill-conditioned.

5 Cross-check.

Extent against minimisation wherever a reaction set exists, because a converged optimisation is not an answer.

6 Carry non-ideality to a measurable.

From $\gamma_{\pm}$ to a cell voltage, and say what ignoring it would have cost.